

MODERN EXPENSE TRACKER

Python Source Code

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import json
import os
from datetime import datetime

DATA_FILE = "expenses.json"

def load_expenses():
    if not os.path.exists(DATA_FILE):
        return []
    try:
        with open(DATA_FILE, "r") as file:
            return json.load(file)
    except json.JSONDecodeError:
        return []

def save_expenses(expenses):
    with open(DATA_FILE, "w") as file:
        json.dump(expenses, file, indent=4)

def add_expense(expenses):
    print("\n--- Add New Expense ---")
    try:
        amount = float(input("Enter amount ($): "))
        if amount <= 0:
            print("■ Amount must be greater than zero.")
            return
    except ValueError:
        print("■ Invalid amount format. Please enter a number.")
        return

    category = (
        input("Enter category (e.g., Food, Transport, Bills, Entertainment): ")
        .strip()
        .capitalize()
    )
    if not category:
        category = "General"

    description = input("Enter description (optional): ").strip()

    new_expense = {
        "amount": amount,
        "category": category,
        "description": description if description else "N/A",
        "date": datetime.now().strftime("%Y-%m-%d %H:%M"),
    }

    expenses.append(new_expense)
    save_expenses(expenses)
    print(f"■ Successfully added ${amount:.2f} under '{category}'.")

def view_expenses(expenses):
    if not expenses:
        print("\n■ No expenses recorded yet.")
        return

    print("\n" + "=" * 65)
    print(f"{'Date & Time':<18} | {'Category':<15} | {'Amount':<10} | {'Description'}")
    print("=" * 65)
    for exp in expenses:
        print(
            f"{exp['date']:<18} | {exp['category']:<15} | "
            f"${exp['amount']:<9.2f} | {exp['description']}"
        )
    print("=" * 65)

def show_summary(expenses):
    if not expenses:
        print("\n■ No data available for summary.")
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    return

total_spent = sum(exp["amount"] for exp in expenses)
category_totals = {}

for exp in expenses:
    cat = exp["category"]
    category_totals[cat] = category_totals.get(cat, 0) + exp["amount"]

print("\n" + "■ EXPENSE SUMMARY " + "■".rjust(25, "="))
print(f"Total Lifetime Spending: ${total_spent:.2f}\n")
print("Breakdown by Category:")
for cat, amt in category_totals.items():
    percentage = (amt / total_spent) * 100
    print(f"    • {cat:<15}: ${amt:>7.2f}  ({percentage:>5.1f}%)" )
print("=" * 45)

def main():
    expenses = load_expenses()

    while True:
        print("\n=== MODERN EXPENSE TRACKER ===")
        print("1. Add Expense")
        print("2. View All Expenses")
        print("3. View Summary & Analytics")
        print("4. Exit")

        choice = input("\nChoose an option (1-4): ").strip()

        if choice == "1":
            add_expense(expenses)
        elif choice == "2":
            view_expenses(expenses)
        elif choice == "3":
            show_summary(expenses)
        elif choice == "4":
            print("\n■ Thank you for using Expense Tracker. Goodbye!")
            break
        else:
            print("■ Invalid choice. Please select a number between 1 and 4.")

if __name__ == "__main__":
    main()

```